Household Spending Patterns in Western Mexico, 2016–2024

Differences by Income, Education, Indigenous Status, and Household Composition

Carlos Torres Orozco
Julio Alberto García Rodríguez
Advanced Topics III: Time Series · October 11, 2026

1 Executive Summary

1.1 Context and problem

The way households in Mexico distribute their living conditions than the total amount they spend. When resources are scarce, spending concentrates on basic needs such as food. When resorces are more abundant, it extends to other goods and services. For this reason, examining the composition of spending helps identify which needs households cover first.

Western Mexico offers an interesting setting in which ti study these patters. The region combines a large metropolitan area, smaller cities, and areas with a large indigenous population.

This report explores how households in Jalisco, Colima, Nayarit and Michoacan distribute their spending on food, clothing, health, transport, aducation and leisure between 2016 and 2024, and how these patterns differ by household composition, income level, indigenous status and the education level of the household head.

1.2 What data were analyzed?

The analysis covers six spending categories:

- Food
- Clothing
- Health
- Transport
- Education
- Leisure

In addition, households were compared along four dimensions: household composition, income level, indigenous status and education level of the household head.

1.3 Relevance for decision-making

Analyzing these data can help inform the scope and tageting of social programs carried out in local communities. It also provides a reference point for observing household spending behavior around the COVID-19 pandemic, which is captured in the 2020 edition.

2 Data

2.1 Source

The data were obtained from the official website of the National Institute of Statistics and Geography (INEGI) and cover the survey editions from 2016 to 2024 (see Appendix A).

2.2 Study region and sample

The region analyzed is Western Mexico, made up of the states of Jalisco, Colima, Nayarit and Michoacan. This region was selected because of its wide variety of urban and rural settings, and the large indigenous communities found in some areas. The sample comprises 47,482 households interviewed accros the four states and five survey editions (2016, 2018, 2020, 2022 and 2024) once expansion factors are applied, each edition represents between 3.9 and 4.7 million households in the region. Table 1 shows the number of sampled households by state and year.

Table 1. Sampled households by state and ENIGH edition.

State	2016	2018	2020	2022	2024
Colima	2,700	2,785	3,282	3,031	3,351
Jalisco	2,017	2,152	2,779	2,625	2,604
Michoacán	1,923	1,874	2,047	2,215	2,130
Nayarit	1,863	1,747	2,103	2,130	2,124
Total sample	8,503	8,558	10,211	10,001	10,209
Expanded households	3,879,364	4,148,619	4,276,550	4,506,307	4,745,110

2.3 Main variables

The main variables fall into six groups: survey design, spending, income, household characteristics, education, and infigenous status. Table 2 lists the variables used in the analysis.

Table 2. Main variables of the analysis

Variable	Type	Description
`factor`, `upm`, `est_dis`	Design	Expansion factor, primary sampling unit, design stratum
`gasto_mon`	Spending	Monetary current spending of the household (quarterly)
`alimentos`, `vesti_calz`, `salud`, `transporte`, `educacion`	Spending	Spending by category
`esparci`, `paq_turist`	Spending	Leisure and tourist packages (leisure = both)
`ing_cor`	Income	Current income of the household (basis of the deciles)
`clase_hog`, `tot_integ`	Household	Household type and number of members
`educa_jefe`	Education	Highest education level of the household head
`hablaind`, `etnia`	Indigenous status	Speaks an indigenous language; self-identifies as indigenous
`parentesco`	Household	Relationship to the household head

2.4 Data preparation and derived variables

Data preparation consisted of removing variables unrelated to household spending and income, standarizing formats and filtering the four Western Mexican states. Derived variables were then build, such as leisure spending, income deciles, household type and indigenous status.

2.5 Limitations

The main limitation is the number of editions, only five editions of the survey are avalibre for this period and each one samples different households, so it is not posible to follow the same household over time. In adition, the share of housheols identified as indigenous drops sharply between 2020 and 2022 in some states, so comparisons by indigenous status are interpreted within each year. Finally, spending is expresed in current pesos of each year, so comparisons across years rely on budget shares rather than amounts.

3 Exploratory Data Analysis

All figures in this section are weighted with the survey expansion factor, so that results describe the population of households in the region rather than only the sampled households.

Spending is expressed as a share of monetary current spending which makes it possible to compare editions without adjusting for inflation, since amounts in pesos are current values of each year. Households reporting no spending were excludes. Unless stated otherwise, comparisons between groups use the 2024 edition, while evolution over time draws on all five editions. The results describe associations between spending patterns and household characteristics, not casual effects, and no confidence intervals were computed.

3.1 Overview of the data

As can be seen in Figure 1, household spending in Western Mexico is skewed: in 2024 the median was 12,538 pesos per month compared with a mean of 17,252, which indicates that a small number of households with very high spending pull the average up. Moreover, not all households spend on the same categories. Food is the most common, followed by transport, while the least common are leisure and education.

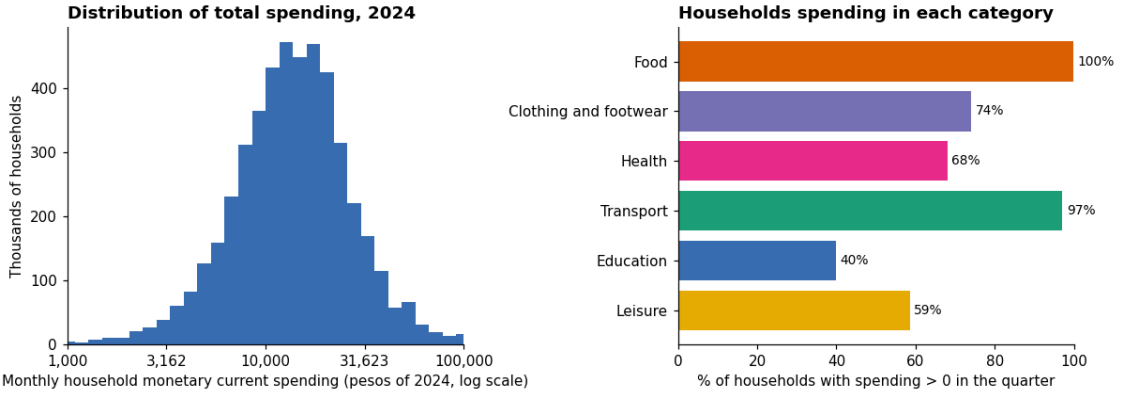


Figure 1. Distribution of monthly household spending and share of households spending in each category, 2024.

3.2 Evolution, 2016–2024

The evolution of household spending between 2016 and 2024 (Figure 2) shows a shift in 2020, a year that coincides with the COVID-19 pandemic. In that year, the share of spending devoted to food rose from 34.1% to 38.2% and the share devoted to health from 3.3% to 4.6%, while the other categories declined, especially

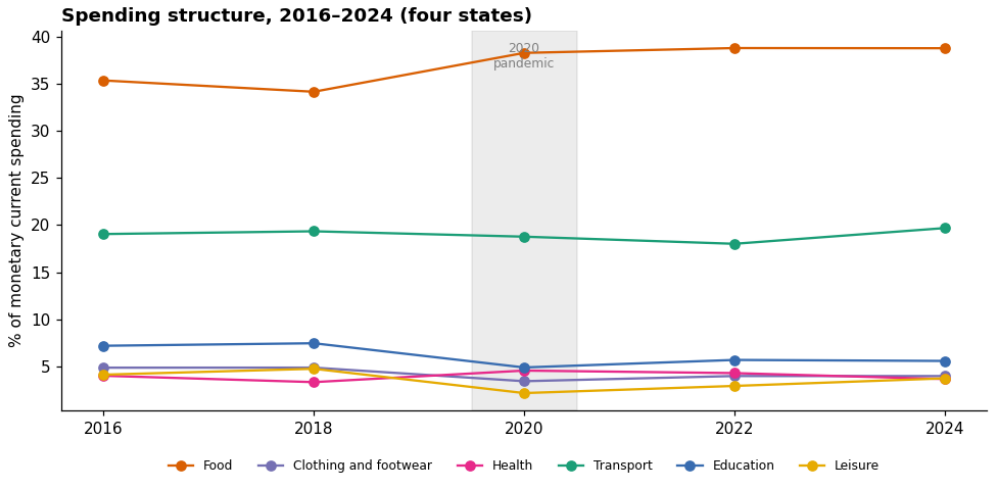


Figure 2. Spending structure by category, 2016–2024.

leisure (from 4.8% to 2.2%) and education (from 7.5% to 4.9%). After 2020 the recovery was only partial: transport and health returned to levels close to those observed before, but food remains high in 2024 (38.7%), and education (5.6%), clothing (4.0%), and leisure (3.7%) remain below their 2016 and 2018 values.

3.3 By income decile

As can be seen in Figure 3, the structure of spending changes markedly with income level. The share devoted to food falls from 49% in the lowest decile to 29% in the highest, while the share devoted to transport rises from 14% to 23%. Education, leisure, and clothing also increase, although more modestly. Health does not follow a clear trend: it fluctuates between 3% and 4% across most of the distribution and reaches 4% in the highest decile.

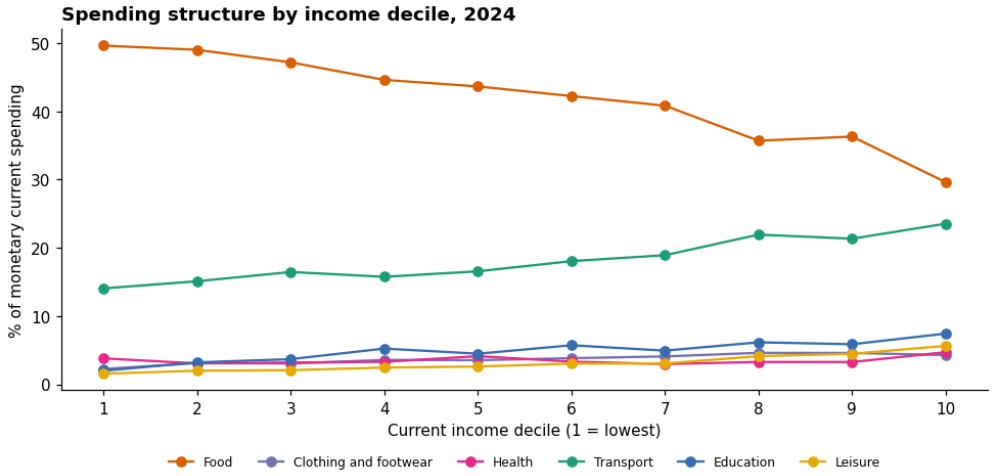


Figure 3. Spending structure by income decile, 2024.

3.4 By household composition

Figure 4 shows differences by household type. Single-person households devote 35% of their spending to food and 20% to transport, and very little to education, but they have the highest share in leisure. Nuclear and

extended households devote a larger share to food and more to education. Per household member, the monthly spending of a single-person household (11,928 pesos) is almost three times that of an extended household (4,151), which suggests economies of scale in larger households. The combined category of composite and co-resident households contains only 114 sampled households, so its figures should be interpreted with caution.

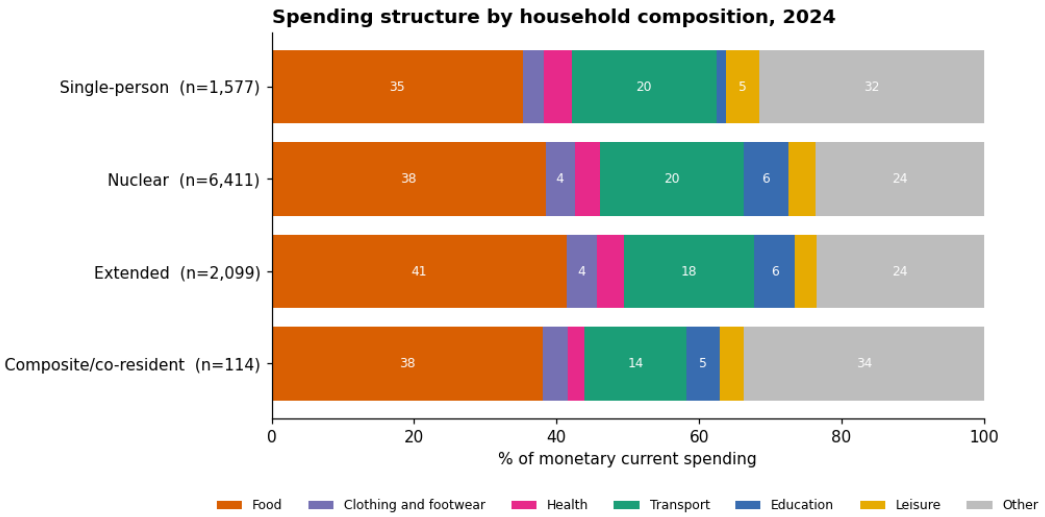


Figure 4. Spending structure by household composition, 2024.

3.5 By education of the household head

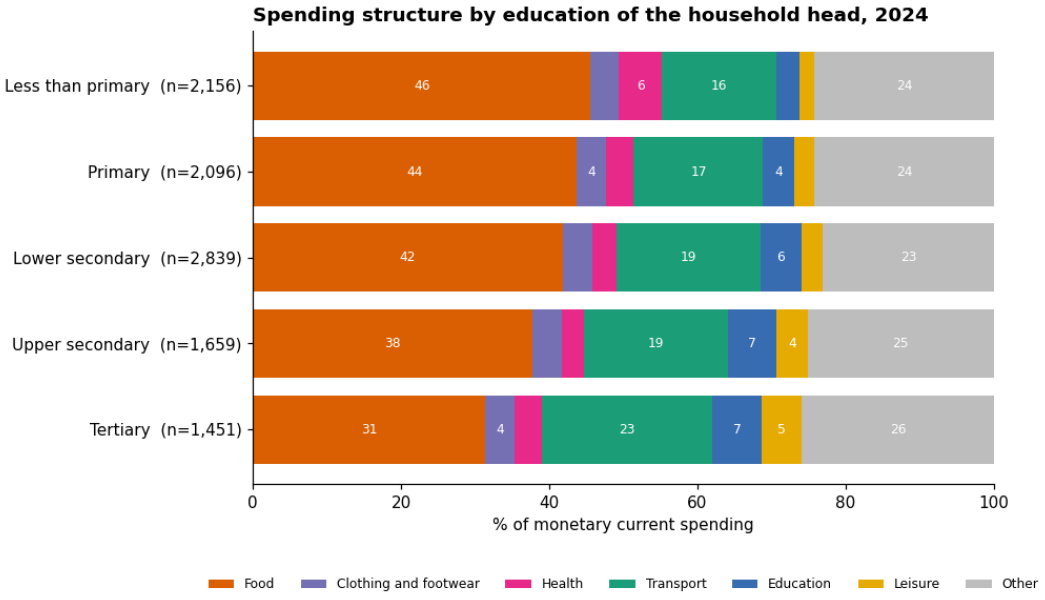


Figure 5. Spending structure by education of the household head, 2024.

As can be seen in Figure 5, the higher the education level of the household head, the lower the share of spending devoted to food and the higher the share devoted to transport, education, and leisure. This pattern resembles the one observed by income level, which is expected because education and income tend to go

together. Health is the exception: households whose head did not complete primary school devote 5% of their spending to it, compared with between 3% and 4% at the other levels, but these household heads are 64 years old on average, compared with between 44 and 47 in the other groups, so the difference mainly reflects age.

3.6 By indigenous status

Figure 6 compares indigenous and non-indigenous households in 2024 (2,060 and 8,141 sampled households, respectively) and shows small differences. The largest are in food and leisure, while the other categories differ by about one percentage point. These differences may reflect household income rather than indigenous status itself, and they are interpreted only within 2024, because the share of households identifying as indigenous changes sharply between 2020 and 2022.

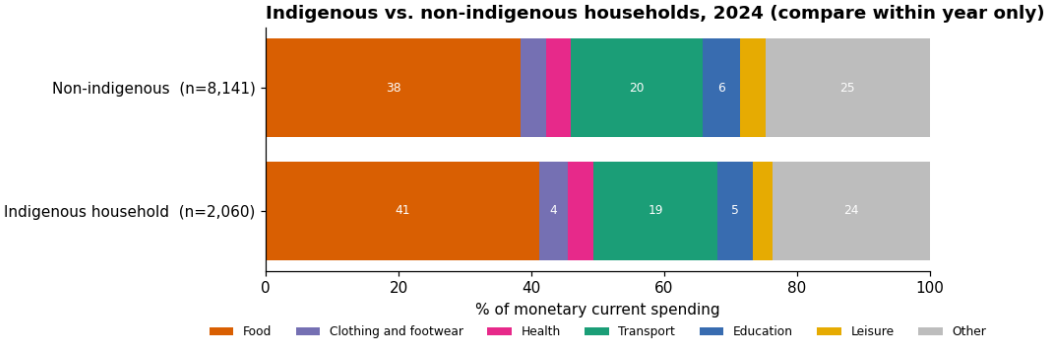


Figure 6. Spending structure of indigenous and non-indigenous households, 2024.

3.7 Persistence across survey editions

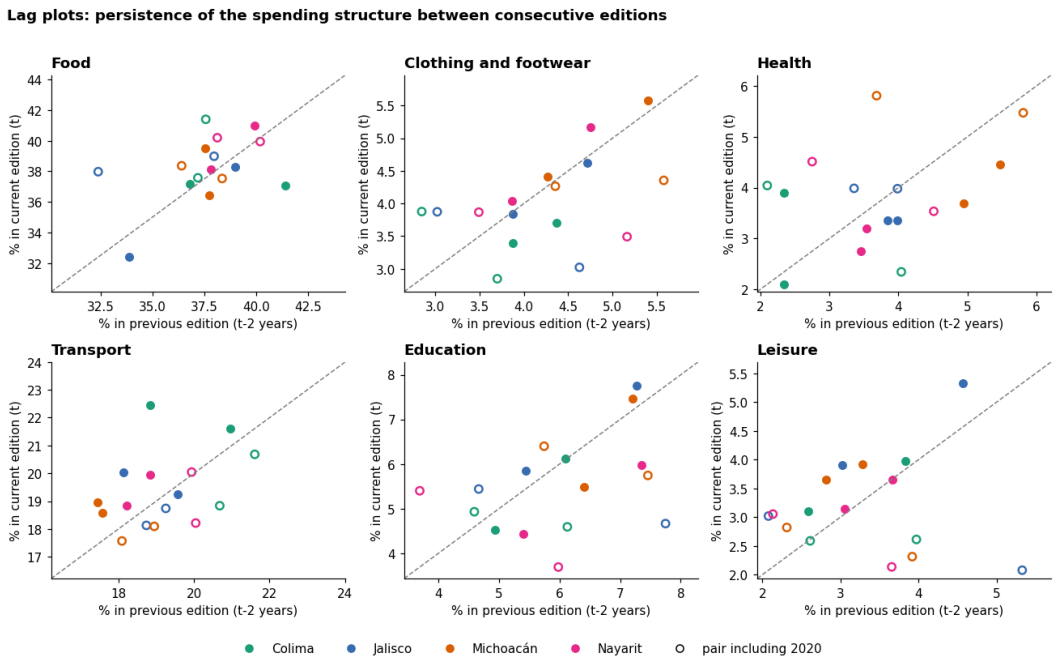


Figure 7. Lag plots of budget shares between consecutive editions (two-year lag), by state.

Figure 7 compares, for each state, the share of spending devoted to each category in one survey edition

against the following edition, with a two-year lag. Most points lie close to the diagonal, which indicates that the structure of spending changes little between consecutive editions: on average, the share of each category changes by between 0.6 and 1.7 percentage points. Food is the most variable category, with an average change of 1.7 points and a maximum of 5.6, while clothing is the most stable, with an average change of 0.6 points. This is a visual check of stability rather than a test of autocorrelation, since only five editions are available.

4 Conclusions

Income level is the factor that most differentiates the spending structure of households in Western Mexico. Moving from the lowest to the highest decile, the share devoted to food falls from 49% to 29%, while the shares of transport, education, and leisure rise (Figure 3). The education level of the household head shows a very similar pattern (Figure 5), which is expected because education and income tend to go together, so this analysis cannot separate the effect of one from the other.

By household type, single-person households devote very little to education, and their monthly spending per member is almost three times that of extended households, which suggests economies of scale (Figure 4). Over time, 2020 marks a shift: the share devoted to food rises from 34% to 38% and remains high in 2024, while leisure and education decline and recover only partially (Figure 2). This shift coincides with the pandemic, but these data do not allow it to be attributed to it. Even so, between consecutive editions the spending structure of each state changes by only 0.6 to 1.7 percentage points on average, so it is relatively stable (Figure 7).

Differences between indigenous and non-indigenous households are small in 2024, but this does not allow us to conclude that no relationship exists: confidence intervals were not computed, the differences may be due to income, and the share of households identifying as indigenous changes sharply between 2020 and 2022, so the comparison is only valid within each year (Figure 6).

The results describe associations, not causal effects. Only five comparable editions of the ENIGH New Series are available (2016 to 2024), each with different households and with spending in current pesos, so forecasting or following the same household over time is not possible. Future work could deflate spending, estimate models that control for income and age simultaneously, compute confidence intervals using the survey design, and incorporate new editions as they are published.

References

[1] Instituto Nacional de Estadística y Geografía (INEGI). *Encuesta Nacional de Ingresos y Gastos de los Hogares (ENIGH)*, New Series, 2016–2024. https://www.inegi.org.mx/programas/enigh/.

A Dataset and repository

All data, scripts, and figures are available at https://https://github.com/Carlos-Torres-Orozco/proyecto-enigh-occidente (version `v1.0`).

Original data. ENIGH, New Series, editions 2016, 2018, 2020, 2022, and 2024, produced by INEGI (https://www.inegi.org.mx/programas/enigh/. Only the `concentradohogar` and `poblacion` tables were used. The original files are not tracked in the repository because of their size; they can be downloaded from INEGI.

Analysis dataset. `data/processed/hogares_2016_2024.csv` contains one row per household and edition (47,482 rows, 50 columns) for Colima, Jalisco, Michoacán, and Nayarit: design variables, spending by category, income, household characteristics, and the derived variables used in the report.

Table 3. Repository contents.

Path	Contents
`02_limpiar.py`	Cleaning: reads the raw INEGI CSV files and filters the four states
`03_unir.py`	Merging and derived variables; writes the analysis dataset
`05_eda.py`	Exploratory analysis: figures and tables
`data/processed/`	Analysis dataset
`figuras/, tablas/`	Figures and tables of the report

Reproduction. Install the dependencies (`pip install -r requirements.txt`) and run `02_limpiar.py`, `03_unir.py`, and `05_eda.py`, in that order.

B Python scripts

All materials used in this report are available in a public GitHub repository (https://https://github.com/Carlos-Torres-Orozco/proyecto-enigh-occidente, version `v1.0`). The repository contains the three Python scripts that reproduce the analysis (`02_limpiar.py` for data cleaning, `03_unir.py` for merging and derived variables, and `05_eda.py` for the exploratory analysis), the analysis dataset (`data/processed/hogares_2016_2024.csv`, with one row per household and survey edition for Colima, Jalisco, Michoacán, and Nayarit), the INEGI data dictionaries and catalogs, and all figures and tables presented in this report. The original ENIGH files are not included in the repository because of their size; they can be downloaded from INEGI (https://www.inegi.org.mx/programas/enigh/). Running the three scripts in the order listed above reproduces the dataset and every figure.